

AskMyPDF — User Manual

Ask a question about a PDF in plain language and get the exact answer back, with the page it came from.

Settings

Model

English (RoBERTa-base)

Run the model

☒ Locally (transformers)

☐ Hugging Face Inference API

> Chunking settings

> OCR (scanned PDFs)

Running on GPU (CUDA)

Model info (Hub API)

[deepset/roberta-base-squad2](#)

Task: question-answering

Downloads (30 days): 496,238

Likes: 961

Language: en

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AskMyPDF

Upload a PDF, ask a question, and an extractive QA model finds the answer span in the document.

Upload a PDF

Upload

200MB per file • PDF

...or try a sample PDF

sample_long.pdf

Pages

3

Words

1,354

Extracted with

pdfplumber

QA mode

chunked (4 ch...

> Preview extracted text

Example questions for this sample

(type my own)

Your question

Which computer defeated Garry Kasparov in 1997?

Get answer

Answer

Deep Blue

Confidence

76.7%

Source

Chunk 3 of 4

Location

pages 2-3

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1. What it does

You give it a PDF and a question. It finds the sentence containing the answer and shows you the exact words, how confident it is, and which page they came from.

It copies text out of your document — it never writes new text. That means it cannot invent a fact, but it also cannot summarise, add up numbers, or answer in its own words. If the document doesn't contain the answer, it says so.

A real example, from the bundled sample:

Q: Which computer defeated Garry Kasparov in 1997? **A:** Deep Blue — 76.7% confident, chunk 3 of 4, pages 2–3

It works in about 100 languages, reads scanned books through OCR, and runs entirely on your machine — nothing is uploaded unless you deliberately choose the hosted option.

2. First-time setup

You need **Python 3.9 or newer**. Open a terminal in the project folder:

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
```

On macOS or Linux the second line is `source .venv/bin/activate`.

That's all. The language model itself (about 500 MB) downloads automatically the first time you ask a question, and is reused from then on — after that the app works offline.

If you only want to read scanned PDFs later, nothing extra is needed now; OCR is already included in `requirements.txt`.

3. Starting the app

```
.venv\Scripts\activate
streamlit run app.py
```

Your browser opens at **http://localhost:8501**. If it doesn't, type that address yourself.

To stop the app, press **Ctrl + C** in the terminal.

4. The screen, explained

Sidebar (left)

Control	What it's for
Model	<i>English</i> is faster; <i>Multilingual</i> understands ~100 languages. Pick Multilingual for any non-English PDF
Run the model	<i>Locally</i> (default, private, no internet needed) or <i>Hugging Face Inference API</i> (runs on Hugging Face's servers, needs a token)
Chunking settings	Leave <i>Auto</i> ticked unless you are experimenting
OCR (scanned PDFs)	Language, page range, and the option to force OCR
 Running on...	Whether it is using your GPU or the CPU
Model info	Live details about the model, fetched from Hugging Face

Main area

1. **Upload a PDF** — or pick one of the bundled samples from the dropdown.
 2. **Four figures** — pages, word count, which library read the file, and whether the document was short enough to read in one go ("direct") or had to be split ("chunked").
 3. **Preview extracted text** — open this to see what the app actually read. Worth a glance whenever an answer looks wrong.
 4. **Your question** and the **Get answer** button.
 5. **The answer**, with confidence, source chunk, page and the surrounding text with the answer highlighted.
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5. Walkthrough A — an English PDF

1. Start the app.
2. Under ...or try a sample PDF, choose **sample_long.pdf**.
3. In **Example questions for this sample**, choose "Which computer defeated Garry Kasparov in 1997?" — or type your own.
4. Click **Get answer**.

You should see **Deep Blue**, about 77% confidence, from pages 2–3, with the sentence shown underneath and the answer highlighted in yellow.

Ask another question without reloading; each answer is added to **Question history**, which you can download as a CSV for your report.

6. Walkthrough B — a non-English PDF

1. Upload your PDF (or pick one from the `testings` folder, e.g. `test_hi_hindi.pdf`).
2. If it isn't in the Latin alphabet, a yellow warning appears with a **Switch to the multilingual model** button — click it. The larger model takes a few seconds to load the first time (and about 2.2 GB on first download).
3. Ask your question **in the same language as the document**, or in English — both work.
4. Click **Get answer**.

The answer comes back in the document's language. Asking "What fruit did Narada bring?" about a Telugu book returns మామిడిపండు, not "a mango". That is what "extractive" means: it hands you the words that are in the document.

Important for Indian languages. Open *Preview extracted text* first. If letters look missing or words are broken (అగ్రా showing as అ), the PDF stores its letters in a way that cannot be read back properly. Fix it by ticking **"Use OCR even if the PDF has text"** in the sidebar — see Walkthrough C. Hindi, Tamil and Punjabi PDFs usually read correctly without this.

7. Walkthrough C — a scanned PDF

A scanned book is a *photograph* of each page. There is no text inside it to search, so the app reads the pages as images instead — this is OCR.

The app spots this by itself:

 **This PDF looks scanned.** Only 10 words were found across 39 pages...

1. In the sidebar under **OCR (scanned PDFs)**, set: - **OCR language** — e.g. *Telugu + English* - **Start at page** — skip covers and contents. If the chapter you want starts on page 5, start there - **How many pages** — start with 5 or 10
2. Tick  **Read this PDF with OCR.**
3. Wait. **OCR takes about 20 seconds per page**, so 6 pages is roughly two minutes. A progress bar shows where it is.
4. Ask your question as usual.

Pages you have already read are remembered, so widening the range later only reads the new pages.

For a PDF that has text but reads badly (the Indian-language case in Walkthrough B), tick **"Use OCR even if the PDF has text"** instead. Example: a Telugu document where the answer came out as అ నగరంలో returns the correct అగ్రా నగరంలో once OCR is on.

8. Reading the results

What you see	What it means
Answer	The exact words from your document
Confidence	How sure the model is, 0–100%. Above ~50% is usually reliable
Source: Chunk 3 of 4	Long documents are read in pieces; this says which piece held the answer
Location: pages 2–3	Where to look in the original PDF
Source context	The passage, with the answer highlighted — <i>always worth checking</i>
 Low confidence	Below 10%. Treat the answer as a guess
Answers from each chunk	What every other part of the document proposed

When it finds nothing

 **No confident answer.** Every chunk preferred "no answer here"...

Two possible reasons:

1. **The document genuinely doesn't say.** This is correct behaviour — the model declining rather than making something up.
2. **It is there but the model missed it.** Look at **Closest guesses** underneath; the right answer is often in that list, just below the model's confidence bar.

If you expected an answer, try: rephrasing the question, checking the extracted-text preview, or switching the model.

9. Settings reference

Model — *English (RoBERTa-base)* is smaller and faster, English only. *Multilingual (XLM-RoBERTa-large)* covers ~100 languages, is about four times larger and slower, and needs roughly 2.5 GB of free memory.

Run the model — *Locally* keeps your document on your machine. *Hugging Face Inference API* sends each question and passage to Hugging Face's servers; it needs a free token in a `.env` file and is rate-limited, but needs no download.

Chunking settings — the model can only read about 512 word-pieces at a time, so long documents are cut into overlapping chunks. *Auto* measures the right size for your document; only untick it if you want to experiment.

OCR — language (one Indian script at a time, plus English), the page range, and the option to use OCR even when the PDF already contains text.

10. Asking good questions

Works well	Works badly
"When was the tower completed?"	"Summarise this document"
"Who built the Taj Mahal?"	"How many times is X mentioned?"
"What BLEU score was achieved?"	"Is this a good paper?"
"Which hardware was used for training?"	"What will happen next year?"

Ask for **a fact that is written in the document**, in the way the document would phrase it. Short, specific questions beat long ones.

11. What it cannot do

- **Summarise, count, calculate or compare.** It only points at existing text.
 - **Join facts from distant pages.** It reads one chunk at a time, so a question needing page 2 *and* page 40 together will fail.
 - **Read handwriting,** or scanned pages in Gujarati, Malayalam, Odia, Punjabi or Tamil (no OCR model available for those).
 - **Read tables reliably.** Numbers in a table arrive as a flat row of digits, so "how many layers does the model have?" often fails on research papers.
 - **Guarantee accuracy on dense technical PDFs.** Expect roughly 4 correct out of 10 on a research paper, against near-perfect on ordinary prose.
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12. Troubleshooting

Problem	Fix
Browser doesn't open	Go to <code>http://localhost:8501</code> manually
"This document isn't in the Latin alphabet..."	Click Switch to the multilingual model
Letters missing, words broken in the preview	Tick Use OCR even if the PDF has text
"This PDF looks scanned"	Tick Read this PDF with OCR and set a page range
"No confident answer" but you know it's there	Check Closest guesses ; rephrase; check the preview
OCR is very slow	Normal: ~20 s per page. Read fewer pages, starting at the one you need
"Could not load the model... too little free memory"	Close other applications, or switch to the English model
"The token is valid but lacks inference permission (403)"	Create a Read token at huggingface.co/settings/tokens and put it in <code>.env</code>
"Rate limit reached"	Switch Run the model back to <i>Locally</i>
First question takes a minute	The model is downloading. Only happens once
<code>KeyError: Unknown task question-answering</code>	<code>pip install "transformers<5"</code>

13. Command reference

Run these with the virtual environment activated.

```
streamlit run app.py                # start the app

python scripts/run_examples.py      # 7 English sample questions
python scripts/run_indic_examples.py # 25 questions in 11 Indian languages
python scripts/run_test_pdfs.py     # the 26 language PDFs in /testings
python scripts/run_test_pdfs.py --ocr # the same, read as images

python -m unittest discover -s tests -v # the test suite (48 tests)
python -m app.hub_info                 # live model details from Hugging Face
```

Sharing a demo as a link — the app accepts URL parameters, so you can open it with a question already answered:

```
http://localhost:8501/?sample=sample_long.pdf&q=Who+coined+the+term+artificial+intelligence
```

For how the system works internally — extraction, chunking, scoring — see the project [README](#). Test results by language are in [testings/README.md](#).